STUDENTS IDENTIFICATION:

| Number: | Name: |
|---|---|
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| 110020 | Diogo Pan |
| 110130 | Diogo Callado |

## 2.1 Simple execution, without data forwarding techniques

**e)**

| | |
|---|---|
| Clock cycles | 308 |
| Instructions | 170 |
| Average CPI | 1.81 |

| | | |
|---|---|---|
| Stalls: | - Data | 104 |
| | - Structural | 0 |
| Flushes: | - Branch Taken | 16 |

**f)** This simulator uses a static predict branch not taken policy. This is evident when looking at the instructions in the pipeline: the final addi and ecall (both come after the branch in the code) are always added, meaning the simulator doesn't change its prediction (branch not taken) even after missing (static).

## 2.2 Application of data forwarding techniques

**c)**

| | |
|---|---|
| Clock cycles | 222 |
| Instructions | 170 |
| Average CPI | 1.31 |

| | | |
|---|---|---|
| Stalls: | - Data | 16 |
| | - Structural | 0 |
| Flushes: | - Branch Taken | 16 |

**d)**

$$\text{speedup} = \frac{CPUT_{old}}{CPUT_{new}} \overset{*}{=} \frac{\frac{CC_{old}}{CR}}{\frac{CC_{new}}{CR}} = \frac{CC_{old}}{CC_{new}} = \frac{308}{222} = 1.38739$$

*assume the clock rate is the same for both setups

## 2.3 Source code optimization: minimization of data and structural hazards

**a)** Attach a copy of the new assembly program.

**c)**

| | |
|---|---|
| Clock cycles | 206 |
| Instructions | 170 |
| Average CPI | 1.21 |

| | | |
|---|---|---|
| Stalls: | - Data | 0 |
| | - Structural | 0 |
| Flushes: | - Branch Taken | 16 |

**d)** Clock rate is assumed to be equal for both setups

$$\text{Speedup} = \frac{CPUT_{old}}{CPUT_{new}} = \frac{CC_{old}}{CC_{new}} = \frac{308}{206} = 1.49515$$

### 2.4 Source code optimization: loop unrolling

**a)** Attach a copy of the new assembly program.

**c)**

| | |
|---|---|
| Clock cycles | 122 |
| Instructions | 110 |
| Average CPI | 1.11 |

| | | |
|---|---|---|
| Stalls: | - Data | 0 |
| | - Structural | 0 |
| Flushes: | - Branch Taken | 4 |

**d)**

Clock rate is assumed to be equal for both setups

$$\text{Speedup} = \frac{CPUT_{old}}{CPUT_{new}} = \frac{CC_{old}}{CC_{new}} = \frac{308}{122} = 2.52459$$

**Table 1:** Pipeline time diagram, without data forwarding techniques.

| | INSTRUCTIONS | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | lw x6 0 x19 | F | D | X | M | W | | | | | | | | | | | | | | F | D | | | | | | | | | | | | | | | | | | | | |
| 2 | lw x7 0 x20 | | F | D | X | M | W | | | | | | | | | | | | | | F | | | | | | | | | | | | | | | | | | | | |
| 3 | mul x28 x7 x6 | | | F | D | – | – | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 4 | add x28 x28 x6 | | | | F | – | – | D | – | – | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 5 | sw x28 0 x27 | | | | | – | – | F | – | – | D | – | – | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | |
| 6 | addi x9 x9 1 | | | | | | | | – | – | F | – | – | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | |
| 7 | addi x19 x19 4 | | | | | | | | | | | – | – | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | |
| 8 | addi x20 x20 4 | | | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | |
| 9 | addi x27 x27 4 | | | | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | |
| 10 | bne x9 x18 -36<loop> | | | | | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | |
| 11 | addi x17 x0 10 | | | | | | | | | | | | | | | | | F | D | | | | | | | | | | | | | | | | | | | | | | |
| 12 | ecall | | | | | | | | | | | | | | | | | | F | | | | | | | | | | | | | | | | | | | | | | |
| 13 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 14 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 15 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 16 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 17 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 18 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 19 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 20 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 21 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 22 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 23 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 24 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 25 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 26 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 27 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 28 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 29 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 30 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

**Table 2:** Pipeline time diagram, with data forwarding techniques.

| | INSTRUCTIONS | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | lw x6 0 x19 | F | D | X | M | W | | | | | | | | | F | D | | | | | | | | | | | | | | | | | | | | | | | | | |
| 2 | lw x7 0 x20 | | F | D | X | M | W | | | | | | | | | F | | | | | | | | | | | | | | | | | | | | | | | | | |
| 3 | mul x28 x7 x6 | | | F | D | - | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 4 | add x28 x28 x6 | | | | F | - | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 5 | sw x28 0 x21 | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 6 | addi x9 x9 1 | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 7 | addi x19 x19 4 | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 8 | addi x20 x20 4 | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 9 | addi x21 x21 4 | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 10 | bne x9 x18 -36<loop> | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | |
| 11 | addi x17 x0 10 | | | | | | | | | | | | F | D | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 12 | ecall | | | | | | | | | | | | | F | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 13 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 14 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 15 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 16 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 17 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 18 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 19 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 20 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 21 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 22 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 23 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 24 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 25 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 26 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 27 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 28 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 29 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 30 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

**Table 3:** Pipeline time diagram, with minimization techniques to reduce the data and structural hazards.

| | INSTRUCTIONS | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | lw x6 0 x19 | F | D | X | M | W | | | | | | | | F | D | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 2 | lw x7 0 x20 | | F | D | X | M | W | | | | | | | | F | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 3 | addi x9 x9 1 | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 4 | addi x19 x19 4 | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 5 | addi x20 x20 4 | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 6 | mul x28 x7 x6 | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 7 | add x28 x28 x6 | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 8 | sw x28 0 x21 | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 9 | addi x21 x21 4 | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 10 | bne x9 x18 -36<loop> | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 11 | addi x17 x0 10 | | | | | | | | | | | F | D | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 12 | ecall | | | | | | | | | | | | F | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 13 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 14 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 15 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 16 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 17 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 18 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 19 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 20 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 21 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 22 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 23 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 24 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 25 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 26 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 27 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 28 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 29 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 30 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

**Table 4:** Pipeline time diagram: usage of loop unrolling minimization techniques to reduce the control hazards.

| | INSTRUCTIONS | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | lw x6 0 x19 | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | F | D | | | | | | | | | | | |
| 2 | lw x7 0 x20 | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | F | | | | | | | | | | | |
| 3 | lw x29 4 x19 | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 4 | lw x30 4 x20 | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 5 | mul x28 x7 x6 | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 6 | add x28 x28 x6 | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 7 | mul x31 x30 x29 | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 8 | add x31 x31 x29 | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 9 | sw x28 0 x21 | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 10 | sw x31 4 x21 | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 11 | lw x6 8 x19 | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | | |
| 12 | lw x7 8 x20 | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | | |
| 13 | lw x29 12 x19 | | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | | |
| 14 | lw x30 12 x20 | | | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | | |
| 15 | mul x28 x7 x6 | | | | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | | |
| 16 | add x28 x28 x6 | | | | | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | | |
| 17 | mul x31 x30 x29 | | | | | | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | | |
| 18 | add x31 x31 x29 | | | | | | | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | | |
| 19 | sw x28 8 x21 | | | | | | | | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | | |
| 20 | sw x31 12 x21 | | | | | | | | | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | | |
| 21 | addi x9 x9 4 | | | | | | | | | | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | | |
| 22 | addi x19 x19 16 | | | | | | | | | | | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | | |
| 23 | addi x20 x20 16 | | | | | | | | | | | | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | | |
| 24 | addi x21 x21 16 | | | | | | | | | | | | | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | | |
| 25 | bne x9 x18 -96 <loop> | | | | | | | | | | | | | | | | | | | | | | | | | F | D | X | M | W | | | | | | | | | | | |
| 26 | addi x17 x0 10 | | | | | | | | | | | | | | | | | | | | | | | | | | F | D | | | | | | | | | | | | | |
| 27 | ecall | | | | | | | | | | | | | | | | | | | | | | | | | | | F | | | | | | | | | | | | | |
| 28 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 29 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 30 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |